Shounak Dey

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PROFESSIONAL SUMMARY

Data Warehouse Engineer with 4+ years of professional experience in end-to-end data pipeline development, PL/SQL programming and ETL operations. Active contributor as a Python developer to open-source projects, including those within the Fedora Infrastructure team.

EXPERIENCE

PROFESSIONAL

BIRLASOFT

Senior Data Engineer — ***Apr 2024 to Present***

Led data integration and performance optimisation initiatives, improving reporting efficiency for business teams

Data Engineer — ***Oct 2022 to Apr 2024***

Designed and optimised database objects and data pipelines, improving performance by 40% and streamlined deployments using Azure DevOps

Trainee Developer — ***Sep 2021 to Oct 2022***

Monitored and troubleshot daily data loads, resolving data integrity issues and improving pipeline stability to ensure 99% on-time data availability

OPEN SOURCE

FEDORA PROJECT

Open-Source Contributor — ***Nov 2024 to Present***

Strengthened mission-critical Fedora Infrastructure services by delivering production-ready features, performance improvements across distributed systems used by millions of users worldwide. Improved schema and event consistency across messaging workflows, optimised data retrieval through materialised views, and elevated software reliability with expanded test coverage and CI automation

General Mentor – Outreachy — ***Mar 2026 to Present***

Mentoring applicants for the Fedora Badges Revamp Project, providing technical guidance on backend architecture and evaluating community contributions

PROJECTS

PERSONAL

LOADOUTS FOR GENSHIN IMPACT

- Co-developed an open-source, cross-platform desktop application using Python, PySide6, and Qt Quick for managing custom equipment configurations
- Integrated Tesseract OCR for automated text extraction from game screenshots, demonstrating ML pipeline implementation
- Achieved 100% test coverage with 1500+ tests across UI and logic layers using Pytest

NOTIFIER FOR GENSHIN IMPACT

- Developed Python-based Telegram bot with dynamic command generation and multi-user support across groups and DMs
- Implemented scheduling algorithms for resource optimisation and personalised user tracking

FASTAPI ECOMMERCE API

- Built fully asynchronous e-commerce REST API with CRUD support for businesses, products, customers, and orders
- Implemented async database operations using SQLAlchemy + asyncpg for PostgreSQL, with Alembic migrations
- Achieved 100% test coverage, ensuring production-ready reliability and scalability

OPEN SOURCE

TAHRIR & TAHRIR-API

- Architected a scalable REST API backend along with delivering full CRUD operations in database API
- Built search features for Fedora's badge platform, improving discoverability and engagement for 40K+ contributors

W2FM & W2FM-MESSAGES

- Expanded Fedora messaging infrastructure capabilities by 25% through GitLab webhook integration, enabling event-driven automation
- Fixed cross-platform event schema bug impacting message consistency across Fedora's distributed infrastructure

PAGURE EXPORTER

- Enabled CentOS infrastructure migration from legacy Pagure to GitLab by delivering secure tag migration capability
- Automated mention sanitisation, enabling safe migration of issues whilst preventing security vulnerabilities

MDAPI

- Prevented race conditions, enhanced data integrity and deployment traceability using atomic file operations with commit-hash versioning
- Improved service flexibility by adding configurable port selection, supporting multi-instance deployments

MIRRORMANAGER

- Architected REST API replacement for legacy XML-RPC check-in system, eliminating deprecated protocols and improving scalability
- Improved user experience and enhanced mirror discoverability by implementing geographic filtering with country-based search

DATANOMMER & FIRMITAS

- Optimised message retrieval performance by implementing materialised database views for high-volume operations in Datanommer
- Elevated code quality standards by achieving 100% test coverage in Firmitas (from 87%)

TECHNICAL SKILLS

Programming Languages	Python, Go, Oracle SQL, PL/SQL
Database	Oracle, MySQL, PostgreSQL
Version Control & CI/CD	Git, GitHub Actions, Azure DevOps
Containerisation	Podman, Docker
Operating Systems	Linux, Windows

ACHIEVEMENT

Birlasoft Team Awesome Award - *for exemplary performance and strong demonstration of customer centricity*

EDUCATION

Bachelor of Engineering (B.E.) in *Electronics & Telecommunication* - Pune University – 2021 | CGPA: 8.37

CERTIFICATIONS

AZ-104 - *Microsoft Azure Administrator Associate*